

Собираем примитивную программу init под ARM:

```
vadim@vadim-PC:~/Emulation/arm-vexpress$ cat init.c
#include <stdio.h>
#include <unistd.h>

int main(void) {
    for (size_t counter = 0;; counter++) {
        printf("%zu\n", counter);

        sleep(1);
    }

    return 0;
}
vadim@vadim-PC:~/Emulation/arm-vexpress$ arm-linux-gnueabi-gcc -static -s init.c -o init
vadim@vadim-PC:~/Emulation/arm-vexpress$ file init
init: ELF 32-bit LSB executable, ARM, EABI5 version 1 (SYSV), statically linked, BuildID[sha1]=474ee66b7362ae06ebfffd16b6e02ca19986d785, for GNU/Linux 3.2.0, stripped
vadim@vadim-PC:~/Emulation/arm-vexpress$
```

Добавляем наш init в cpio-архив и сжимаем при помощи gzip:

```
vadim@vadim-PC:~/Emulation/arm-vexpress$ echo init | cpio -o -H newc | gzip
> initramfs.cpio.gz
659 blocks
vadim@vadim-PC:~/Emulation/arm-vexpress$ file initramfs.cpio.gz
initramfs.cpio.gz: gzip compressed data, from Unix, original size modulo 2^32 337408
vadim@vadim-PC:~/Emulation/arm-vexpress$
```

Теперь у нас есть (примитивная) корневая файловая система, попробуем запустить новую прошивку:

```
vadim@vadim-PC:~/Emulation/arm-vexpress$ QEMU_AUDIO_DRV=none qemu-system-arm
-M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-c
a9.dtb -append "console=ttyAMA0" -nographic
[    0.000000] Booting Linux on physical CPU 0x0
[    0.000000] Linux version 6.18.33 (vadim@vadim-PC) (arm-linux-gnueabi-g
cc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, GNU ld (GNU Binutils for Ubuntu)
2.42) #1 SMP Fri May 29 15:29:39 +07 2026
[    0.000000] CPU: ARMv7 Processor [410fc090] revision 0 (ARMv7), cr=10c538
7d
[    0.000000] CPU: DIT / VIRT nonaliasing data cache, VIRT nonaliasing inc
```

```
[ 1.872954] PM: genpd: Disabling unused power domains
[ 1.911442] Freeing unused kernel image (initmem) memory: 2048K
[ 1.914725] Run /init as init process
[ 1.939974] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/b
us@40000000/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-b
us@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input1
0
[ 2.060635] input: AT Raw Set 2 keyboard as /devices/platform/bus@4000000
0/bus@40000000:motherboard-bus@40000000/bus@40000000:motherboard-bus@4000000
0:iofpga@7,00000000/10006000.kmi/serio0/input/input2
1
2
3
4
5
6
7
8
9
[ 11.991451] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 11.991748] amba 100e0000.memory-controller: deferred probe pending: (rea
son unknown)
[ 11.991803] amba 100e1000.memory-controller: deferred probe pending: (rea
son unknown)
[ 11.991834] amba 100e5000.watchdog: deferred probe pending: (reason unkno
wn)
10
```